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SUPERSEDING
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MILITARY SPECIFICATION

BOOTS, DUST AND WATER SEAL (FOR TOGGLE AND PUSH-BUTTON SWITCHES, CIRCUIT BREAKERS, AND ROTARY-ACTUATED PARTS), GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for molded silicone rubber boots which can be used on component parts such as toggle and pushbutton switches, circuit breakers, and rotary-actuated parts such as rotary switches, variable resistors, capacitors, inductors, and transformers. The boots protect the component parts-actuating mechanism from sand, dust, water and other contaminants, and seal the panel on which the component parts are mounted.

1.2 Classification. The boots covered by this specification shall be one of the types and styles as shown below. The applicable information shall be as shown on the individual specification sheet (see 3.1).

Type A - For panel mounted toggle switches and circuit breakers.

Style 1 - One-hole mounting, enclosed.

Style 3 - One-hole mounting, through.

Style 4 - Three-hole mounting, through.

Style 6 - One-hole mounting, miniature, enclosed.

Style 7 - One-hole mounting, subminiature, enclosed.

Style 8 - One-hole mounting, subminiature, through.

Style 9 - Five-hole mounting, through.

Style 10 - Panel bushing seal only.

Type B - For bushing mounted pushbutton switches.

Style 1 - One-hole mounting adapter type.

Style 2 - One-hole mounting drilled panel.

Style 3 - One-hole mounting through.

Style 4 - One-hole mounting miniature, drilled panel.

Style 5 - Three-hole mounting (for telephone handset switches).

Type C - For bushing mounted rotary actuated parts.

Type D - For use as seals for lever operated circuit breakers.

Style 1 - Panel mounted without mounting frame.

Style 2 - Panel mounted complete with mounting frame.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Electronics Research and Development Command, ATTN: DRDEL-ED, Adelphi, MD 20783 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMEMTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-N-290 - Nickel Plating (Electrodeposited).

QQ-P-416 - Plating, Cadmium (Electrodeposited).

ZZ-R-765 - Rubber, Silicone.

MILITARY

MIL-G-7703 - Guard, Switch.

MIL-S-28786 - Switches, Preparation for Delivery of.

(See Supplement 1 for list of associated specification sheets.)

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.

MIL-STD-1285 - Marking of Electrical and Electronic Parts.

MIL-STD-45662 - Calibration Systems Requirements.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMEMTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern (see 6.2).

3.2 Qualification. Boots furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.5 and 6.3).

3.3 Materials The materials shall be as specified herein. However, when definite materials are not specified, materials shall be used which will enable the boots to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product (see 4.3)

3.3.1 Silicone rubber Boots shall be molded of silicone rubber in accordance with ZZ-R-765, class 3a, grades 25, 50, or 60 as specified (see 3.1). Unless otherwise specified (see 6.2), the color shall be gray.

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3.3.2 Metals. Metals shall be of corrosion resistant material, or shall be plated to resist corrosion. Plating shall be in accordance with class 2, (0.0003 inch nickel) of QQ-N-290, or class 2, type II of QQ-P-416.

3.3.3 Lubricant. Sealing portions of the boot which come in intimate contact with the actuating lever or shaft of the component part on which it is mounted, shall be lubricated as specified (see 3.1) with a transparent permanent lubricant. The lubricant shall not cause deleterious effects in the physical properties of the rubber, i.e., deterioration of the rubber, embrittlement, softening or swelling of the rubber, and shall in no manner prevent the boots from complying with any other requirements of this specification.

3.4 Design and construction. Boots shall be of the design, construction, and physical dimensions specified (see 3.1). For boots having metal inserts, design of the insert shall provide for mechanical (metal-to-panel) stop against the panel to which the boot is mounted. The purpose of the stop is to prevent over-tightening of the boot and to avoid damage to the sealing member. Unless otherwise specified, all threaded parts shall be in accordance with FED-STD-H28.

3.5 Restoring force (torque) (applicable only to type A boots without shielding). When boots are tested in accordance with 4.7.2, the force required to restore the boot to either extreme position shall not exceed 5 inch-ounces unless otherwise specified (see 3.1). The time to complete the switch actuator movement a minimum of 20 degrees shall not exceed 5 seconds in either direction.

3.6 Watertightness. When boots are tested as specified in 4.7.3 through 4.7.3.2, as applicable, there shall be no evidence of water leakage into the test enclosure, or a continuous stream of air bubbles through or around the component-part mounting bushing, if applicable.

3.7 Durability.

3.7.1 Operation of guards (applicable only to type A styles used with hinged-type guards (see 3.1). When boots are tested as specified in 4.7.4.1, type A boots suitable for use with hinged-type guards, such as those in accordance with MIL-G-7703, shall not interfere with the satisfactory operation of the guard. There shall be no evidence of damage to the boot.

3.7.2 Installation. When boots are tested as specified in 4.7.4.2, there shall be no evidence of tearing of the boot.

3.8 Sand and dust. When boots are tested as specified in 4.7.5, there shall be no evidence of entrance of sand or dust from the front of the switch panel.

3.9 Vibration. When boots are tested as specified in 4.7.6, there shall be no evidence of loosening of the boot.

3.10 Salt spray (corrosion) (applicable only to boots with threaded inserts or metal frames. As a result of the test specified in 4.7.7, there shall be no evidence of excessive corrosion. Excessive corrosion shall be construed as any corrosion which exposes the base metal.

3.11 Endurance. When boots are tested as specified in 4.7.8 through 4.7.8.3, there shall be no evidence of brittleness or cracking. The boot shall not cause the switch actuator to bind or malfunction during the test.

3.12 Lubricant. When tested as specified in 4.7.9.1, the lubricant shall show no signs of flow, creep, or spread, nor shall the lubricant sink into the rubber thicken, evaporate, or leave a dry, opaque film or residue. The lubricant shall not have congealed to a point where it no longer functions as a lubricant when tested as specified in 4.7.9.2.

3.13 RFI shielding (applicable only to boots having an RFI shielding rating - "S"). When boots are tested in accordance 4.7.10, the resistance value shall not exceed 1 ohm before the endurance test and 1.5 ohms following the endurance test.

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3.14 Marking. The unit package shall be marked in accordance with method I of MIL-STD-1285. The marking shall include as a minimum.

a. Military part number.

b. Manufacturer's name, trademark, or federal supply code for manufacturers.

c. Date code.

3.14.1 Military part number. The military part number shall consist of the letter "M", the basic number of the specification sheet, an assigned dash number (see 3.1).

M	5423/07	-06
Military designator	Specification sheet number	Dash number

3.15 Workmanship. Boots shall be processed in such a manner as to permit them to meet all the requirements of this specification. Flash shall be kept to a minimum consistent with good molding practice, and in no case shall rubber prevent the proper installation of the boot on the applicable switch or test plug.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-STD-45662

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows

a. Materials inspection (see 4.3).

b. Qualification inspection (see 4.5)

c. Quality conformance inspection (see 4.6).

4.3 Materials inspection. Materials inspection shall consist of certification, supported by verifying data, that the materials listed in table I that are used in fabricating the boots are in accordance with the applicable referenced specifications or requirements prior to such fabrication (see 3.3).

TABLE I. Materials inspection.

Material	Requirement paragraph	Applicable specification
Silicone rubber	3.3.1	ZZ-R-765
Metals/plating	3.3.2	QQ-N-290 or QQ-P-416

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4.4 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.4.1 Test plugs. The installation, sand and dust, vibration, endurance, and watertightness inspections of type C boots shall be made while the boots are mounted on test plugs which shall be assembled from interchangeable shafts and bushings in accordance with figure 1. The details of shafts and bushings to be used in the assembly of the applicable test plugs shall be as specified (see 3.1). Fifty percent of the boots shall be tested with the applicable maximum shaft-size test plug and fifty percent of the boots shall be tested with the applicable minimum shaft-size test plug.

TABLE II. Qualification inspection (non-RFI shielded boots).

Inspection	Requirement paragraph	Method paragraph
Group I - All sample units (10)		
Visual and mechanical inspection	3.1, 3.3, 3.4, 3.14, 3.15	4.7.1
Restoring force (type A only)	3.5	4.7.2
Watertightness (preliminary)	3.6	4.7.3.1
Group II - 6 sample units		
Durability		
Operation of guards 1/	3.7.1	4.7.4.1
Installation	3.7.2	4.7.4.2
Sand and dust	3.8	4.7.5
Group III - 4 sample units		
Vibration	3.9	4.7.6
Salt spray (corrosion) 2/	3.10	4.7.7
Group IV - All sample units		
Endurance	3.11	4.7.8 through 4.7.8.3
Watertightness (final)	3.6	4.7.3.2
Group V 3/		
Lubricant	3.12	4.7.9

1/ Applicable only to type A boot styles used with hinged-type guards.

2/ Applicable only to boots with metal inserts or metal frames.

3/ Test to be conducted on sample slab.

4.5 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.3) on sample units produced with equipment and procedures normally used in production (see 3.2).

4.5.1 Sample size. Ten sample boots of each military part number shall be subjected to qualification inspection as specified in table II or IIa, as applicable.

4.5.2 Test slab. A 6-inch by 6-inch by 0.075-inch thick slab of rubber as the same grade (see 3.1 and 3.3.1) as the boots under test shall be submitted for the lubricant test.

4.5.3 Failures. One or more failures shall be cause for refusal to grant qualification approval.

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TABLE IIa. Qualification inspection (RFI shielded boots).

Inspection	Requirement paragraph	Method paragraph
Group I - All sample units (10)		
Visual and mechanical inspection	3.1, 3.3, 3.4, 3.14, 3.15	4.7.1
Watertightness (preliminary)	3.6	4.7.3.1
Group II - 6 sample units		
Installation	3.7.2	4.7.4.2
Sand and dust	3.8	4.7.5
RFI shielding (3 samples only) only)	3.13	4.7.10
Group III - 4 sample units		
Vibration	3.9	4.7.6
Salt spray (corrosion) 1/	3.10	4.7.7
Group IV - 7 sample units 2/		
Endurance	3.11	4.7.8 through 4.7.8.3
Watertightness (final)	3.6	4.7.3.2
RFI shielding	3.13	4.7.10
Group V 3/		
Lubricant 4/	3.12	4.7.9

1/ Applicable only to boots with metal inserts or metal frames.

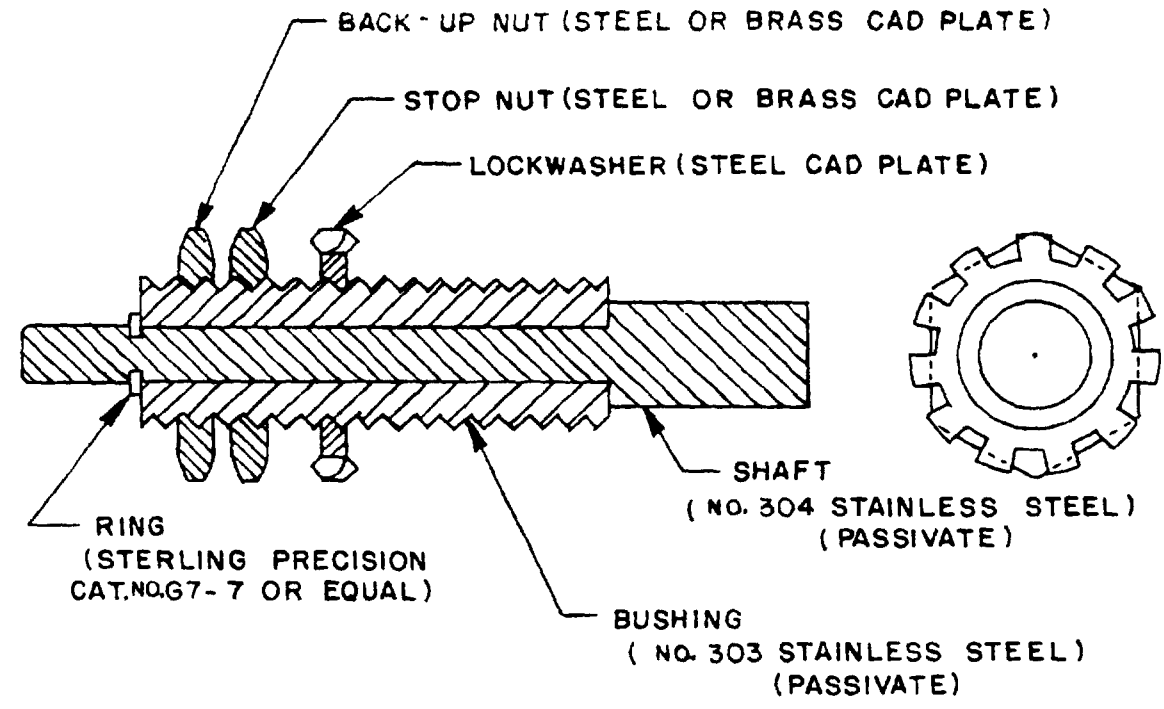
2/ Samples previously subjected to RFI shielding test shall not be included in group IV.

3/ Test to be conducted on sample slab.

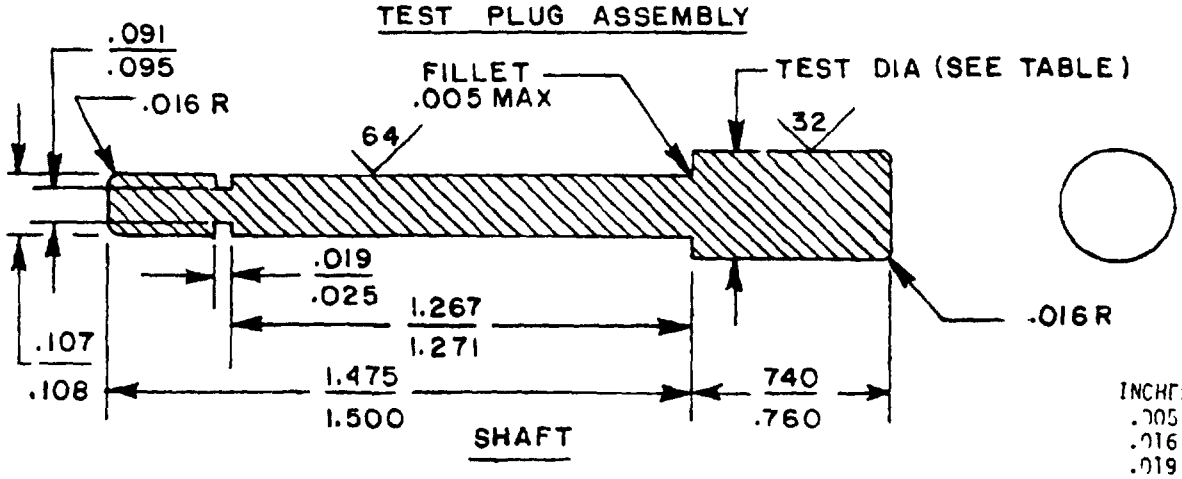
4/ Test not required if conducted under table II.

4.5.4 Retention of qualification. To retain qualification, the contractor shall forward to the qualifying activity at 12-month intervals, a summary of the results of groups A and B inspections performed during that period, indicating as a minimum the number of lots that have passed, the number of lots that have failed, and the complete results of group C inspections, including the number and type of any part failures. When the inspection data submitted for the first 12-month interval indicates conformance to specification requirements, subsequent group C inspections shall be extended to a 24-month interval. The contractor shall forward for each 24-month interval the complete results of group C inspections performed during the last 12 months of that interval. If the inspection results indicate nonconformance with specification requirements, action may be taken to remove the failing product from the qualified products list. Failure to submit the summaries and inspection data shall result in loss of qualification for that product. In addition to the periodic submission of inspection data, the contractor shall immediately notify the qualifying activity anytime during the applicable interval that the inspection data indicates failure of the product to meet the requirements of this specification. In the event that no production occurred during the 12-month reporting interval, a report shall be submitted certifying that the manufacturer still has the capabilities and facilities necessary to produce the item If the item has not been produced for 24 months, the product must be requalified.

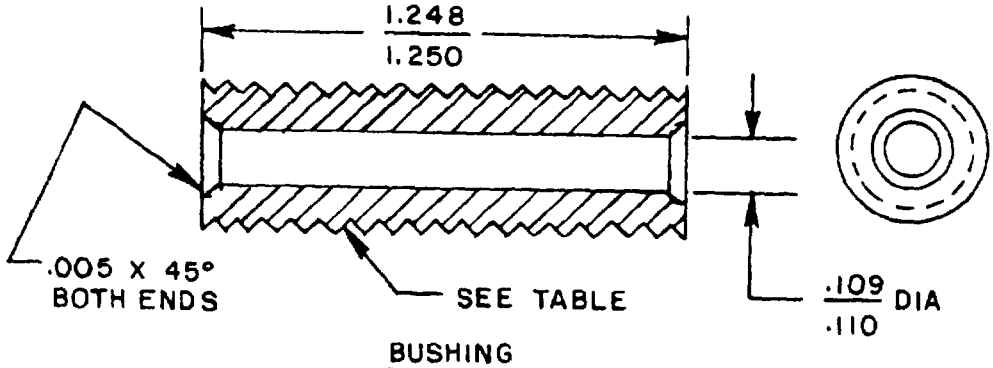
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TEST PLUG ASSEMBLY



SHAFT



BUSHING

INCHES	MM
.005	.13
.016	.41
.019	.48
.025	.64
.091	2.31
.095	2.41
.107	2.72
.108	2.74
.109	2.77
.110	2.79
.740	18.80
.760	19.30
1.248	31.70
1.250	31.75
1.267	32.18
1.271	32.28
1.475	37.47
1.500	38.10

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Test plugs for inspection of type C boots.

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SHAFT DETAILS

Shaft letter	Shaft size	Shaft test diameter, + .001
A	.125 Min	0.121
B	.125 Max	0.128
C	.250 Min	0.246
D	.250 Max	0.253
E	375 Min	0.371
F	.375 Max	0.378

BUSHING DETAILS

Bushing No.	Thread
1	.250 -32 NEF-2A
2	.375 -32 NEF-2A
3	.469 -32 NS-2A
4	.500 -32 NS-2A
5	.625 -24 NEF-2A

INCHES	MM
.001	.03
.121	3.07
.125	3.18
.250	6.35
.375	9.53
.469	11.91
.500	12.70

FIGURE 1 Test plugs for inspection of type C boots - Continued.

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4.6 Quality conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspections.

4.6.1.1 Inspection lot. An inspection lot shall consist of all of the boots of the same type, style, and size produced under essentially the same conditions and offered for inspection at one time.

4.6.1.2 Group A inspection. Group A inspection shall consist of the inspections specified in table III, in the order shown.

4.6.1.2.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality level (AQL) shall be as specified in table III. Major and minor defects shall be as defined in MIL-STD-105.

TABLE III. Group A inspection.

Inspection	Requirement paragraph	Method paragraph	Defect classification	AQL (percent defective)	
				Major	Minor
Visual and mechanical inspection	3.1, 3.3, 3.4	4.7.1	Major	1.0	4.0
Marking	3.14		Minor		
Workmanship	3.15		Major		

4.6.1.2.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects or screen out the defective units and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots and shall be clearly identified as reinspected lots.

4.6.1.2.3 Disposition of sample units. Sample units which have passed all the group A inspection shall be delivered on a contract or purchase order if the lot is accepted.

4.6.1.3 Group B inspection. Group B inspection shall consist of the inspections specified in table IV in the order shown, and the sample shall be selected from inspection lots that have passed group A inspection.

TABLE IV. Group B inspection.

Inspection	Requirement paragraph	Method paragraph
Restoring force (type A only) 1/ 3/	3.5	4.7.2
Durability (installation)	3.7.2	4.7.4.2
Watertightness (preliminary)	3.6	4.7.3.1
RFI shielding 2/ 3/	3.13	4.7.10

1/ Not required for RFI shielded boots.

2/ Required for RFI boots only.

3/ Only 4 sample boots from each inspection lot are required.

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4.6.1.3.1 Sampling plan. The sampling plan shall be in accordance with MIL-STD-105 for small-sample inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. For small samples, reduced inspection procedure, R-1, shall be used. The AQL shall be 6.5 (percent defective), and the inspection level shall be S-4 for normal and tightened inspection, and S-3 for reduced inspection.

4.6.1.3.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects or screen out the defective units and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots and shall be clearly identified as reinspected lots.

4.6.1.3.3 Disposition of sample units. Sample units which have passed all of the group B inspection shall be delivered on a contract or purchase order if the lot is accepted; however, RFI shielded boots that are of enclosed style (type A or type B) and constructed with the RFI shield molded within the boot shall not be delivered.

4.6.2 Periodic inspection. Periodic inspection shall consist of group C inspection. Except where the results of these inspections show noncompliance with the applicable requirements (see 4.6.2.1.3), delivery of products which have passed groups A and B inspections shall not be delayed pending the results of these periodic inspections.

TABLE V. Group C inspection.

Inspection	Requirement paragraph	Method paragraph
(All sample units)		
Durability (operation of guards) 1/	3.7.1	4.7.4.1
Sand and dust	3.8	4.7.5
Vibration	3.9	4.7.6
Salt spray (corrosion) 2/	3.10	4.7.7
Endurance	3.11	4.7.8 through 4.7.8.3
Watertightness (final)	3.6	4.7.3.2
Lubricant 3/	3.12	4.7.9

1/ Applicable only to type A boots sytles used with hinged-type guards.

2/ Applicable only to boots with metal inserts or metal frames.

3/ Test to be conducted on sample slab.

4.6.2.1 Group C inspection. Group C inspection shall consist of the inspections specified in table V in the order shown. Group C inspection shall be made on sample units selected from inspection lots which have passed the groups A and B inspections.

4.6.2.1.1 Sampling plan. Ten sample units shall be selected from those covered by a single type, style, and size, 12 months after the date of notification of qualification, and after each subsequent 24-month period. If group C inspection is to be performed on RFI shielded boots of the same type, style, and size at the same time, the sample size may be split equally between the shielded and nonshielded boots. The contractor shall also submit a 6- by 6- by 0.075 inch thick slab of rubber of the same grade (see 3.1 and 3.3.1) as the boots under test, for the lubricant test. The manufacturer is also required to submit verification that the materials used in his samples are in accordance with the applicable specification requirements (see 4.3).

4.6.2.1.2 Disposition of sample units. Sample units which have been subjected to group C inspection shall not be delivered on a contract or purchase order.

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4.6.2.1.3 Noncompliance. If a sample fails to pass group C inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product that can be corrected and that were manufactured under essentially the same conditions, with essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the qualifying activity). Groups A and B inspections may be reinstituted, however, final acceptance and shipment shall be withheld until the group C inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

4.6.3 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of MIL-S-28786.

4.7 Methods of inspection.

4.7.1 Visual and mechanical inspection. Boots shall be examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements and that there is lubricant present in the applicable portions of the boots (see 3.1, 3.3, 3.4, 3.14 and 3.15.)

4.7.2 Restoring force (torque) (applicable only to type A boots). The restoring force shall be measured using equipment similar to that shown on figure 2 (see 3.5).

4.7.2.1 Procedure. The following details shall apply.

a. Sample boots shall not have been flexed or operated for at least 24 hours prior to the test.

b. With the latching device in place so that the opeating arm cannot move, the boot shall be mounted on an appropriate toggle actuator. Normal switch mounting means shall be used to secure the boot to the actuator.

c. The 1 ounce weight shall be mounted to the left side of the operating arm in the 5-inch position.

d. The latch shall be released and the time measured for the switch to complete a 20-degree movement (minimum) as shown on the dial.

e. With the boot still mounted, the arm shall be returned to horizontal position and the latching device secured. The 1-ounce weight shall be mounted to the right side of the operating arm in the 5-inch position.

f. The latch shall be released and the time measured for the switch to complete a 20-degree movement (minimum) as shown on the dial.

4.7.3 Watertightness.

4.7.3.1 Preliminary. Boots shall be mounted on the applicable switches, dummy switches, circuit breakers, or test plugs, (all without an internal bushing seal) (see 4.4.1) and installed on a test enclosure. Boots shall be submerged in water to a depth of 6 ±2 inches and subjected to an increasing internal pressure at a rate of 1 pound force per square inch gage ($lbf/in^2(gage)$) per minute, until a pressure of 15 pounds-force per square inch gage is reached, at which time the switches, circuit breakers, or test plugs shall be operated for twenty-five cycles. The boots shall again be submerged in water to a depth of 6 ±2 inches and subjected to an increasing external pressure at a rate of 1 pound-force per square inch gage per minute, until a pressure of 15 pound-force per square inch gage, unless otherwise specified (see 3.1), is reached, at which time the switches or test plugs, as applicable, shall again be operated for twenty-five cycles. Cycles of operation shall be as specified in 4.7.8. During the test, boots shall be observed for evidence of water leakage into the test enclosure, or a continuous stream of air bubbles (see 3.6).

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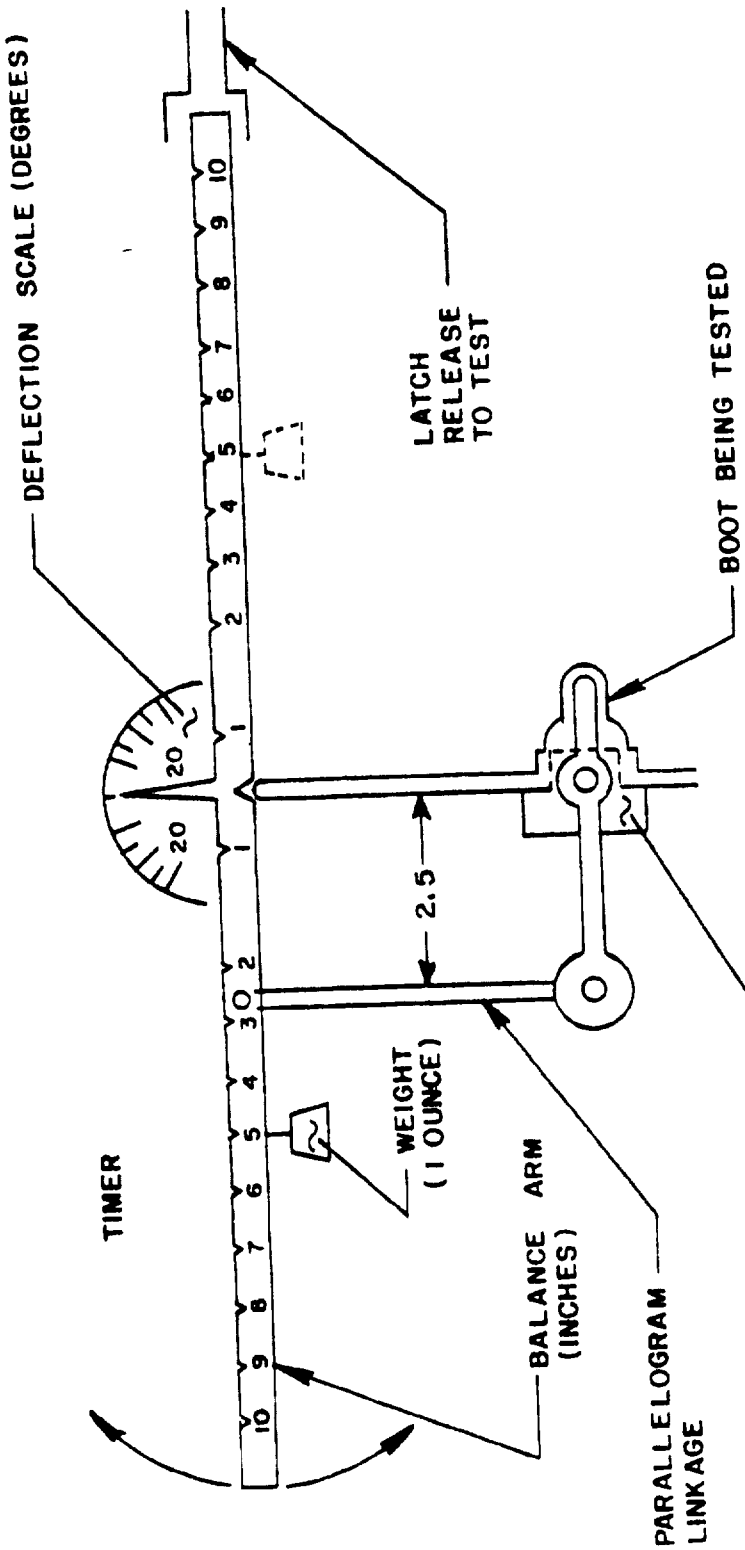


FIGURE 2. Restoring force test fixture.
(Schematic)

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4.7.3.2 Final. Boots shall be tested as specified in 4.7.3.1, except that the internal pressure shall be increased at a rate of 1 pound-force per square inch gage every two minutes until a pressure of 15 pounds-force per square inch gage is reached, and maintained at that pressure for 30 minutes. The external pressure shall be similarly applied, except that the pressure of 15 pounds-force per square inch gage, unless otherwise specified (see 3.1), shall be maintained for 1 1/2 hours. During these periods of maximum pressure, the switches or test plugs, as applicable, shall be operated for twenty-five cycles at the maximum internal pressure and for twenty-five cycles at the maximum external pressure. Cycles of operation shall be as specified in 4.7.8. During the test, boots shall be observed for evidence of water leakage into the test enclosure, or a continuous stream of air bubbles (see 3.6). For boots with an inner constriction feature (see 3.1) one-half of the samples for qualification and group C inspections shall have the rubber portion removed just above the inner constriction before performing the test.

4.7.4 Durability.

4.7.4.1 Operation of guards (applicable only to type A styles with hinged-type guards). Boots shall be mounted on a panel, together with an appropriate hinged-type guard and appropriate switch. The guard shall be opened and closed at a minimum rate of ten operations per minute. For qualification inspection, the number of operations shall be five hundred, and for group C inspection, the number of operations shall be fifty. No restraining force shall be applied to the guard during the closing operation. During the test, there shall be no interference with the satisfactory operation of the guard. After the test, boots shall be examined for evidence of damage (see 3.7.1).

4.7.4.2 Installation. Boots shall be installed and removed from an applicable switch, circuit breaker, or test plugs (see 4.4.1) fifty times during qualification inspection and five times during group B inspection. After the test, boots shall be examined for evidence of tearing (see 3.7.2).

4.7.5 Sand and dust (see 3.8). Boots shall be tested in accordance with method 110 of MIL-STD-202. The following details shall apply:

a. Mounting conditions: Applicable switches without an internal bushing seal, applicable circuit breakers, or applicable test plugs (see 4.4.1) shall be installed on a test enclosure comprising a sealed container, with only the front of the switch, circuit breaker, or test plug exposed to the environment, prior to mounting of the boot.

b. Operation: During steps 1 and 2 the applicable switches, circuit breakers, or test plugs, with mounted boots, shall be subjected to 2,500 cycles of operation (see 4.7.8 at a rate of 10 to 12 cycles per minute (cpm).

c. Measurements: Not applicable.

After the test, the test enclosure shall be examined for evidence of sand and dust.

4.7.6 Vibration (see 3.9). Boots shall be tested in accordance with method 201 of MIL-STD-202. The following details shall apply:

a. Method of mounting Boots shall be mounted on applicable switches, circuit breakers, or applicable test plugs (see 4.4.1), which, in turn, shall be mounted on a stainless steel panel.

b. Measurements: Not applicable.

After the test, boots shall be examined for evidence of loosening.

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4.7.7 Salt spray (corrosion) (applicable only to boots with threaded inserts or exposed metal frames) (see 3.10). Boots shall be tested in accordance with method 101 of MIL-STD-202. The following details shall apply.

a. Special mounting Boots shall be suspended from the top of the chamber.

b. Test condition letter B.

c. Measurements after exposure Not applicable.

After the test, boots shall be examined for evidence of excessive corrosion of the base metal.

4.7.8 Endurance. Boots shall be mounted on applicable switches, circuit breakers, or test plugs (see 4.4.1) and subjected to the cycles of operation and temperature specified in 4.7.8.1 through 4.7.8.3. For types A, B, and D boots, a cycle of operation shall be a movement of the switch or circuit breaker actuator from one extreme position to the other extreme position, and return to the original position. For type A boots, a minimum angle of excursion of 30° (15° each side of center) shall be applied. For type C boots, a cycle of operation shall be one 360 ±5 degree-movement of the rotating shaft in a clockwise direction, and return 360 ±5 degrees in the counterclockwise direction. For qualification inspection, cycling shall be at a rate of 10 to 12 cpm. Faster cycling during group C inspection is optional. During the test, there shall be no binding or malfunctioning of the switch actuator. After the test, boots shall be examined for evidence of brittleness or cracking (see 3.11).

4.7.8.1 At room ambient temperature. Boots shall be subjected to 30,000 cycles of mechanical operation (see 4.7.8), unless otherwise specified (see 3.1).

4.7.8.2 At -65° ±2°C. The temperature of the test chamber shall be maintained at -65° ±2°C for 24 hours. During the last 2 hours, boots shall be subjected to 1,000 cycles of mechanical operation (see 4.7.8).

4.7.8.3 At 85° ±5°C. The temperature of the test chamber shall be maintained at 85° ±5°C for 24 hours. During the last 2 hours, boots shall be subjected to 1,000 cycles of mechanical operation (see 4.7.8).

4.7.9 Lubricant (see 3.12).

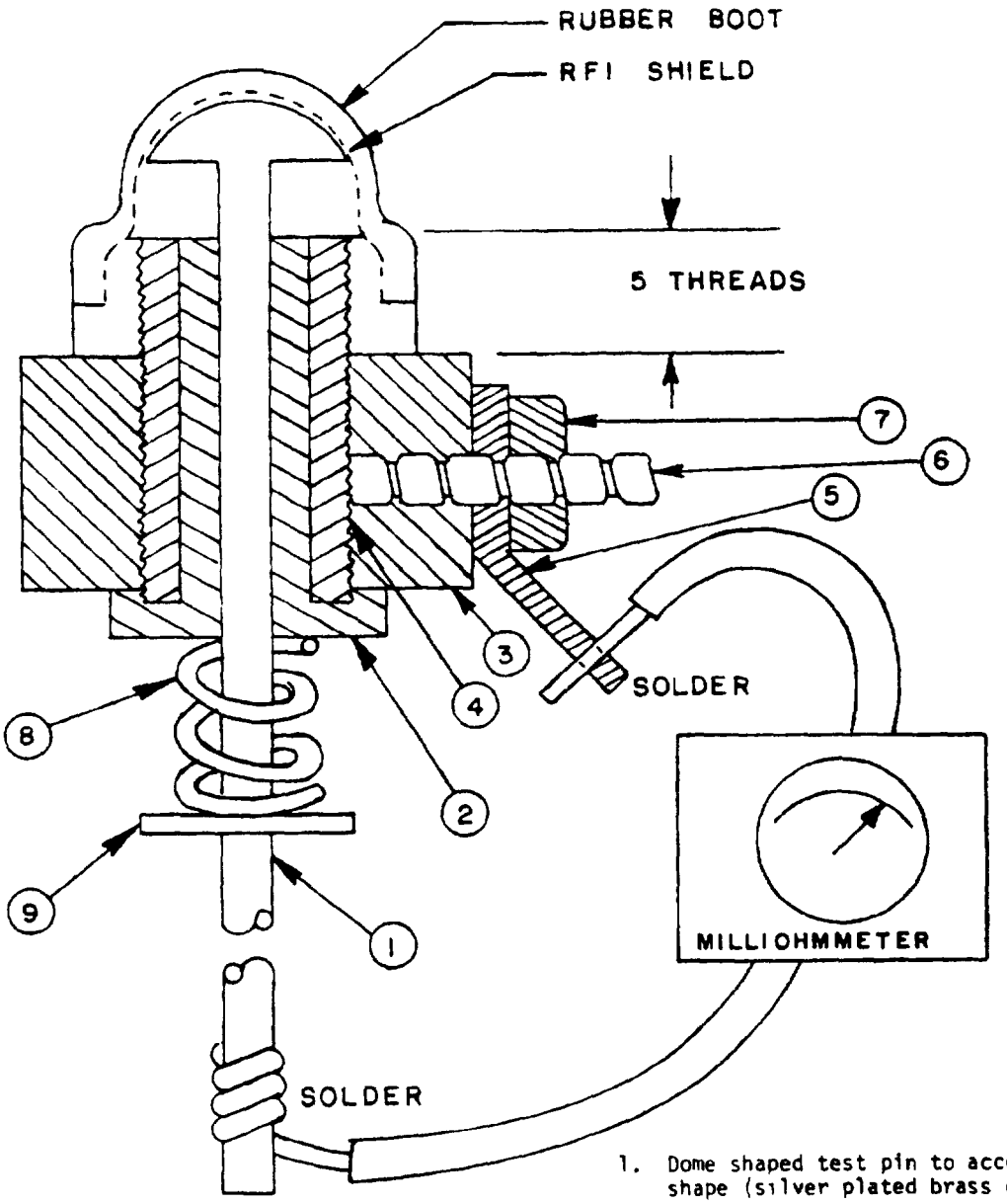
4.7.9.1 At high temperature. Three stripes of lubricant, each approximately 1/4-inch by 1-inch shall be applied to a 6- by 6- by 0.075-inch slab of rubber of the same grade (see 3.1 and 3.3.1) as the boots under test. The rubber slab shall be subjected to a temperature of 255° ±5°C for 4 hours, and then allowed to cool to room ambient temperature for 1 hour. The rubber slab shall be examined for evidence of flow, creep, or spread of the stripes of lubricant, sinking of the lubricant into the rubber, thickening or evaporation of the lubricant, presence of a dry, opaque film or residue at the stripes on the rubber slab where the lubricant was applied, or deleterious effects on the rubber. The lubricant shall be transparent at the end of the test.

4.7.9.2 At low temperature. After completion of the test specified in 4.7.9.1, the rubber slab shall be subjected to a temperature of -65° ±2°C for 1 hour. The stripes of lubricant shall then immediately be manipulated with a test stick to determine whether the lubricant has congealed to the point where it no longer functions as a lubricant.

4.7.10 RFI shielding (applicable only to boots having an RFI shielding rating - "S") (See 3.13)

4.7.10.1 For types A and B enclosed style boots with nonmolded RFI mesh and for boots using conductive rubber (see 3.1). The test sample boot shall be assembled to the test fixture using a minimum of 5 threads (as shown on figure 3) The spring shall force the dome shaped test pin against the internal mesh (for nonmolded type)

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1. Dome shaped test pin to accommodate boot shape (silver plated brass or equal)
2. Insulator - Polystyrene
3. Barrel - 1 inch dia drill rod.
4. Bushing - brass
5. Lug
6. Set screw
7. Nut - brass
8. Spring - to exert upward force on test pin of 3/4 to 1 pound.
9. Retaining ring.

FIGURE 3. RFI test fixture (typical) for enclosed, nonintegral conductive rubber boots.

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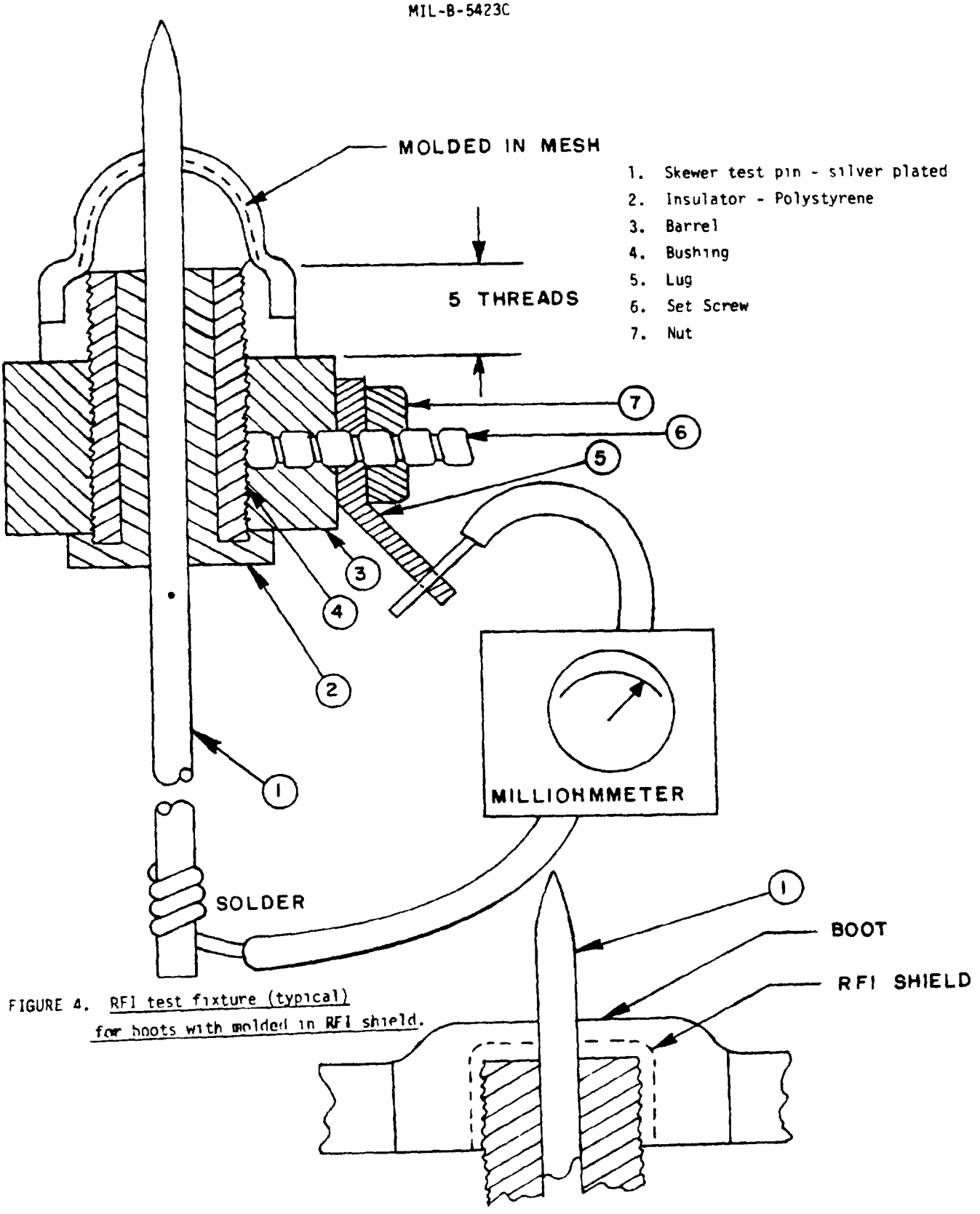


FIGURE 4. RFI test fixture (typical) for boots with molded in RFI shield.

FIGURE 4a. RFI test fixture (typical) for boots with open construction.

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or against the boot interior (for conductive rubber boots). The shape of the test pin shall accommodate the boot shape and size. The maximum current used during the measurement of resistance, using a milliohmmeter (or other equivalent method) shall be 100 milliamperes. The resistance between the test pin and the connection lug shall be measured. The values of resistance measured shall not exceed that specified by 3.13.

4.7.10.2 For boots with molded RFI mesh (see 3.1). The procedures and equipment shall be similar to that described in 4.7.11.1 except that a skewer pin (see figure 4) is used in place of the domed test pin. The spring is not required. The skewer is forced upward into the boot until significant penetration is accomplished. The balance of the test procedure is the same as 4.7.10.1. The values of resistance measured shall not exceed that specified by 3.13.

4.7.10.3 For type A, style 3, 4, 8, and 9 and type C boots (see 3.1). The test procedure of 4.7.10.2 shall be used except that for type C boots the test pin diameter shall be identical to that of the shaft of the switch the boot will accommodate. For type A through style boots, the pin diameter shall be as specified (see 3.1). The balance of the procedure shall be identical to that for 4.7.10.2. The values of resistance measured shall not exceed that specified by 3.13.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-28786.

6. NOTES

6.1 Intended use. Boots covered by this specification are intended to provide protection for panel mounted components against such environments as sand, dust, water and other contaminants. The boots seal the shaft or other external parts as well as seal the panel on which the component parts, such as rotary switches, toggle switches, pushbuttons, circuit breakers, etc., are mounted.

6.2 Ordering data. Acquisition documents should specify the following.

a. Title, number, and date of this specification.

b. Title, number, and date of the applicable specification sheet and the complete military part number (see 3.1).

c. Color, if other than gray (see 3.3.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable qualified products list (QPL), whether or not such products have actually been so listed by that date. The attention of the contractors is called to this requirement, and manufacturers are urged to arrange to have products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the qualified products list is the U.S. Army Electronics Research and Development Command, Ft. Monmouth, NJ 07703, however, information pertaining to qualification of products may be obtained from the Defense Electronics Supply Center (DESC-E), Engineering Standardization Directorate, Dayton, OH 45444.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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